

Signage

6.1 Overview

A new signage system for NAVSTA Great Lakes will have a significant presence on the site, bringing clarity through a unified set of signs. The signage will help bring order to both vehicular and pedestrian traffic by providing clear directions and information to visitors, staff, and recruits. Signage should be strategically located at key decision points and should include the minimum amount of detail to avoid visual clutter.

The similarities between signage systems for NAVSTA Great Lakes and the two other Installations within the Region will provide for ease of upgrades and maintenance. These guidelines outline the tasks required to accomplish an integrated, coordinated signage system, which includes the programming of messages and locations, as well as the design of the actual elements. The design principles and major sign types are described in each IAP and are intended to be extrapolated into other elements as the system is designed and expanded.



Examples of existing signage at Great Lakes



6.1.1 Image

A well-designed signage system can instill a strong element of image and identity. A signage system should utilize consistent materials and detailing to be a unifying site amenity and improve wayfinding. This system should apply to the entry gate and continue through the site to visitors' destinations.

Elements of the signage design which contribute to a positive image and identity include:

- Consistency of design in styling, message layouts and information
- Use of logos, details and graphic elements including typefaces
- Styling, including colors and design elements, in keeping with the buildings and architecture of the base

6.1.2 Programming

Programming is the process of determining the exact message and location for each specific sign throughout the project. Programming is typically a major effort, requiring extensive client input. Representative examples of each sign type are provided; these samples can be adapted to specific conditions defined during programming.



Examples of existing signage at Great Lakes

MESSAGES

Messages on signs should be limited to make signage comprehensible and usable. Determining sign messages is an exercise in compromise because the large number of destinations and users on the property is at conflict with the need to limit messages to maintain clarity.

The Navy can facilitate the sign upgrade process by establishing an internal decision-making process and authority to work with a signage designer to create an inventory and signage message schedule listing each sign with its messages and directional arrows.

LOCATIONS

The programming process will determine and document the actual location of each sign on a signage location plan, found at the end of this chapter. Each sign is labeled to coordinate with the message schedule, and building elevations, photos, or details are provided as appropriate to convey clearly each sign location. Particular attention is given to signs within the primary streets or "VIP Routes" that would be prominent during a tour of the Installation.

The most basic criterion for determining sign location is function, particularly for directional or identification signs, such as wayfinding signs



placed at critical decision-making points. Other factors include visibility, access for service, and integration with other site and building elements. In addition, illuminated signs must have an accessible power source.

At each Installation, each proposed sign location must be reviewed for visibility, existing obstacles, and available space. The following guidelines will further ensure legibility and consistency:

- Directional signs should only be posted at the right side of the road, so that drivers will be able to anticipate sign locations and easily view the signs. This will also reduce visual clutter from sign backs.
- Similarly, avoid posting signs back-to-back to enhance legibility and predictability of sign locations.

NOMENCLATURE

A successful sign system relies on a consistency and clarity of nomenclature. Destinations at an Installation should always be referred to with identical language. A sign copy wording list, defining



Avoid posting signs back-to-back

the wording to be used for each destination and direction, should be developed to ensure consistency and to reduce the possibility of confusion.

- In general, destinations should have short, functional names. For example, DISPENSARY or THE NAVY MUSEUM are better building names than THE ADMIRAL JOHN Q. DOE BUILDING because they are easier to remember and provide information about their function.
- Only the most universally understood acronyms should be used. Where possible, spell out the names of buildings and destinations. Remember that the intended audience for the signs may be unfamiliar with obscure terminology. This is especially true for visitor and tourist destinations.
- Short messages are better. For example, Slow instead of Drive Slowly, or Men instead of Gentlemen.
- Provide as much information as possible. For example, Vertical Clearance 8 Feet instead of Caution! Low Overhead.
- State messages positively. Say Staff Only instead of No Visitors.

6.1.3 Maintenance

Signs are not maintenance-free. A review of existing signs on the Installations reveals a variety of conditions, from good to in need of maintenance or cleaning due to vehicular damage; fading, flaking paint; dirt and soot; and peeling vinyl messages. At best, materials typically specified in sign systems last between three and ten years before requiring replacement or major maintenance. An updated system must include an ongoing budget and assign responsible parties to maintain it.



However, steps can be taken to increase longevity and minimize maintenance. These include:

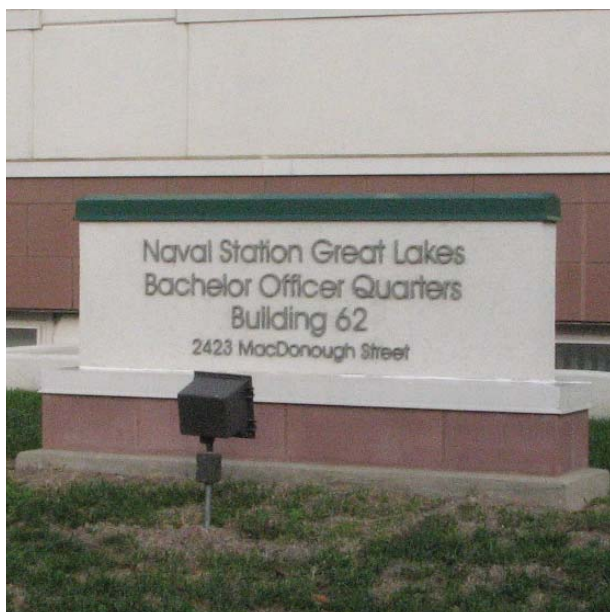
- Locating signs out of vehicular and pedestrian contact zones
- Providing a landscape barrier to prevent damage from mowing and snow-removal operations
- Specifying high-quality finishes, such as anodized metals, porcelain enamel, and automotive-quality paint finishes
- Designing for planned message changes, to prevent damage and inconsistency arising from field-applied message

Some maintenance is inevitable. Tasks include:

- Cleaning
- Periodic inspection, damage repairs and finish touch-up

- Message changes, replacements and repairs. Replacing messages requires investment in specialized tools, as well as a sensitivity to typography and the design goals of the system.

These tasks may be divided between in-house sources and out-sourced service contracts with sign vendors, but should not be left to chance. It is important that the design goals of the system be maintained over time. Strong consideration should be given to what tasks are best done by a signage specialist. Experience shows that in-house repairs and message changes by maintenance crews often compromise the design goals of sign systems by using different materials, inconsistent typefaces, and poor spacing.



Examples of existing signage, demonstrating inconsistencies with message design and layout

6.2 Sign Components

It is important to provide information in a simple and clean layout, with small amounts of text to enhance clarity of the message. The components of a sign that require careful coordination include message layout, including color, size and spacing, number of messages, and message order; typefaces; and hierarchy of sign types.

6.2.1 Sign Hierarchy

In order to program, locate, and design the signage system, a hierarchy of elements must be developed as part of a sequence. This sequence begins when a visitor first encounters graphics at the entrances accessed from surrounding public streets. From there, it progresses into the site, to drop-off and parking locations, and on to specific building, office, or visitor destinations. To leave, the sequence must be repeated in reverse. It is important to provide information in small amounts and not load the graphics with unnecessary messages so that decisions can be made at appropriate intersections.

A variety of sign types are required to provide a sign hierarchy. Signs that are intended to direct vehicular traffic are different from those oriented to pedestrians; they have a larger font size and a smaller number of messages. However, these distinct types of signs should include common themes of colors, type styles, and message wording in their design to create a cohesive, useful system. Once the system is in place, it will be a significant amenity to the site through its clarity through organization of sign types.



Signage hierarchy - from most important at the top to least at the bottom

6.2.2 Message Layout

COLOR

In general, light text on a dark background is easier to read and allows for the use of reflective text on roadway signage. The background color for all signs on the Installation shall be blue, Pantone Color #19-4052 TPX, as dictated by the standards set in the Navy Installation Appearance Guidelines (IAG). For the historic district, sign background color should be Pantone 19-1625 TPX Decadent Chocolate with white reflective lettering.



The letter color for all signage shall be white.
Each sign shall also display a color band that corresponds to the functional district in which the sign is located.

FUNCTIONAL DISTRICT COLOR BAND

All signage will display a color band located above the signage field that will be colored according to the functional district in which the sign is located. These functional districts will be defined as existing functional districts become further developed and occupy distinct areas of the Installation. These future districts will be generalized to the most common activity type in that area. In the case that a building's function does not match the activity defined by its district, the signage for that building will reflect the district, not the building's individual function.

Pantone 19-4110 TPX: Midnight Navy

Pantone 19-1625 TPX: Decadent Chocolate

Pantone 12-0424 TPX: Light Gray

Pantone 15-4101 TPX: High Rise

Pantone 17-4405 TPX: Monument

Color will be used as a signage feature to complement the identity of the site, as well as to differentiate between zones of the Installation.

Administrative and Training
Myrtle

Pantone # 18-6114 TPX

Personnel Support
Rattan

Pantone # 14-1031 TPX

Industrial
Monument

Pantone # 17-4405 TPX

Waterfront

Bluesteel

Pantone # 18-4222 TPX



Color will be used as a signage feature to complement the identity of the site, as well as to differentiate between zones of the Installation.

The Functional District color band scheme is intended to aid a sailor on an unfamiliar Installation in identifying his whereabouts so he or she may quickly and easily reach his destination. In the case of NAVFAC Great Lakes, all signs would be color-coded for Administrative and Training, Personnel Support, Industrial, and Waterfront.

SIZE AND SPACING

Messages should be sized and spaced for visibility from their intended users. Vehicular and pedestrian signs have very different text size requirements, which are described in those specific sections. Travel speed, sign color, contrast, illumination, typeface, message spacing, and other variables can also affect the legibility of messages.

At least half the letter height is required to separate lines of text; where different font sizes are used on adjacent lines, separation equalling half the height of the larger font is required.

NUMBER OF MESSAGES

Six messages is the maximum that can be comprehended by drivers in a moving vehicle. Pedestrian signs can accommodate more messages, since users can linger and read. Directories should be designed and placed for use by pedestrians.

ABCDEFGHIJKLMN
OPQRSTUVWXYZ

abcdefghijklmn
opqrstuvwxyz

1234567890

Sample font: Palatino Linotype

MESSAGE ORDER

Destinations should be listed in the order that they will be encountered. The closest destination should be at the top of the list. As a specific destination is approached, it moves up the message hierarchy.

6.2.3 Typefaces

Typefaces or lettering should be standardized to ensure uniformity and comprehension of the information conveyed on the sign.

- The typeface(s) designated for a site should be used on all signs, with the exception of any words or lettering that is part of a specific logo or other unique graphic element.
- Lettering should be both upper and lower case, not all capitals.
- No more than two type sizes should be displayed on any one sign.
- Type sizes should always remain constant for each line of type.
- Numbers should not be spelled out unless they are part of a recognized acronym.

ABCDEFGHIJKLMN
OPQRSTUVWXYZ

abcdefghijklmn
opqrstuvwxyz

1234567890

Sample font: Helvetica Medium



- The lettering and background colors must have a strong contrast.
- Helvetica Medium should be the predominant typestyle for messages. Helvetica Bold may be used when a hierarchy of lines of information is needed. Helvetica Condensed should only be used when the intended line of type will not otherwise fit on the sign.
- Within the historic district, building numbers should be Palatino Linotype, nine inches high, white in color. Sub-headlines in the historic district should consist of Helvetica Medium or Helvetica Bold when a hierarchy of lines of information is needed.
- For signs intended to be viewed from vehicles, lettering should have a lower case letter height in accordance with the signed speed limit, as given in the following table.

Speed Limit (mph)	Lower Case Letter Height
<25	3"
25-35	4"
40 or 45	6"

Vehicular design speeds and their associate signage font sizes

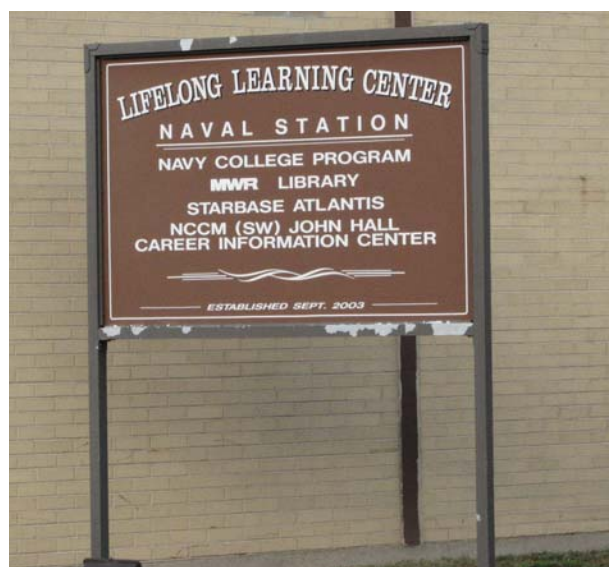


6.3 Directional Signage

Directional signage guides visitors around a facility, as they enter the site and move through the Installation by vehicle or on foot. The legibility and position of directional signs, as well as the order of information and the content provided on the sign, is critical for an easily read and functional signage system. Directional signage should be free-standing, never posted on buildings or other structures. Particular types of directional signage are given below.

6.3.1 Site Entry Identifiers

The Site Entry Identifiers located upon entry of the project perimeter are the starting point of the visitor experience and create a valuable first impression. They identify the Installation from the surrounding roads and streets and form an important symbolic role as the “gateway” to the Installation. These signs should introduce the graphic theme of the site to the visitor, to provide visual cues on what to look for to guide them through the site.



Examples of existing signage, demonstrating inconsistencies with message design and layout



The following design features apply to the signage used on site entry identifiers:

- The Site Entry Identifiers should introduce the project signage color scheme, typeface, and logo to the visitor. In particular, materials and color schemes should match those identified for Site Building Identification Signs.
- The Navy Seal shall be incorporated within each gate entry sign.
- The design should be carefully coordinated and themed with the architectural and landscape features. The sign itself may be combined with these elements by mounting on architectural or landscape walls, integrating with planting, and using compatible materials.

- The function of Site Entry Identifiers should not be combined with directional messages; they should be kept solely as an entry marker.



Current signage is incorporated into the gate concept; there are no stand-alone site identifiers.



Recommended options for site entry identification signs. The bottom option reflects existing signage for the Historic District



- Primary and secondary entrances to the sites should have similar sign treatments at different scales. The same basic design should be adapted at a smaller, appropriate size for secondary entrances.
- Clear direction to visitor check-in locations should be provided at or before entry to each gate, as applicable.
- Locations should be chosen to maximize impact and visibility.
- Signs can be illuminated in order to be visible at night and in low light. Uplighting or any lighting placement that creates light spill should be avoided (Refer to Lighting: Minimizing Light Pollution, Section 7.6.9 for additional information).

6.3.2 Street Name Identification Signs

The image of the Street Identification Signs should be cohesive, by using the project sign color scheme and consistent typography. The street identification signs may differ by district, if appropriate.

- The following design features apply to street identification signs:
- Street name signs should be placed prominently at all intersections.
- Street name identification signs may use different signage elements like colored bands to distinguish districts or other place names from street names.



Existing street name identification signs.



Existing street name identification signs.



Existing street name identification sign in the historic district. The Ross Field identification panel conflicts with street name identification

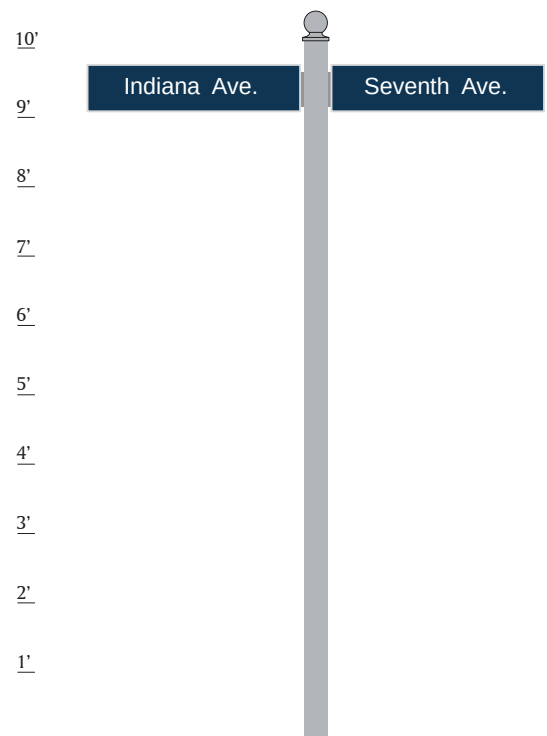
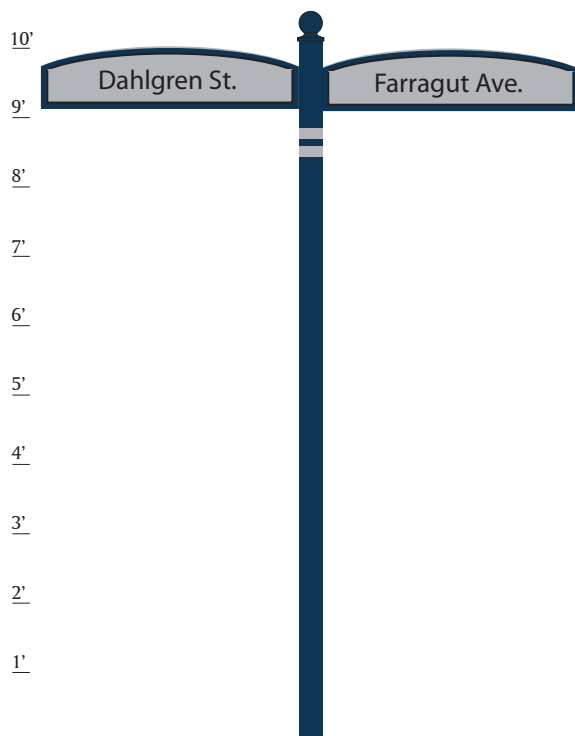
6.3.3 Vehicular Directional Signs

Vehicular Directional Signs provide information for those traveling by car. They are key to facilitating vehicular movement to appropriate parking and drop-off locations.

Primary directional signs are located in key areas such as near Installation entries, primary streets (VIP routes), and major intersections or decision-making points. They should be larger and more prominent than secondary directional signs located along general routes. Primary signs should be the most prominent signs on the Installation, second only to signage at entry gates.

The following design features apply to vehicular directional signs:

- Messages should direct vehicles to parking and drop-off locations. Directions to individual buildings and functions are not appropriate if there are no parking facilities available.
- Care should be taken in placing signs to avoid blocking drivers' vision at intersections. As few signs as possible should be installed to avoid visual clutter.
- Messages should be sized to be visible from vehicles at the appropriate speed. Refer to the Typeface guidelines given above for text sizes.
- Messages should be placed high enough to avoid being blocked by vegetation and other cars, as appropriate. All messages should be a minimum of 30" above grade for design speeds of 30 mph or below and 36" above grade for design speeds above 30 mph.



Recommended street name identification signs for historic district (left) and typical configuration (right)



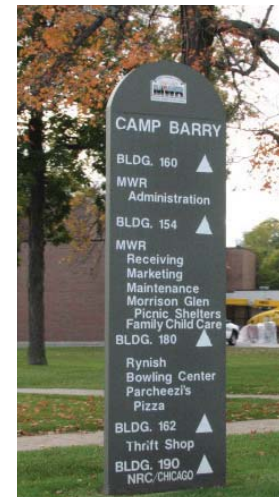
- The exiting sequence should be signed, to provide directions to the surrounding streets.
- Avoid listing building numbers on vehicular directional signs, where possible. Where necessary to differentiate similarly named buildings, the building number shall be justified to the right edge of the sign.

PRIMARY DIRECTIONAL SIGNS

- Primary signs should contain the facility name as the top message panel.
- Consider illuminating primary directional signs, either internally or by floodlights. Uplighting or placement of fixtures that creates light spill should be avoided (Refer to Lighting: Minimizing Light Pollution, Section 7.6.9 for additional information).
- If illumination is cost prohibitive, reflective vinyl lettering should be used.

SECONDARY DIRECTIONAL SIGNS

- Secondary directional signs should utilize forms and materials similar to the primary signs.



Existing vehicular directional signs displaying inappropriate solutions.



Recommended vehicular directional sign

8'
7'
6'
5'
4'
3'
2'
1'

- Drivers can comprehend a maximum of six messages on a sign from a moving vehicle. More messages will cause drivers to slow down or stop to review the sign. Consider limiting messages or using additional signs instead.
- White text on a dark background is most legible. Reflective vinyl should be used on vehicular directional signs.
- Signs shall be constructed of heavy-gauge aluminum with automotive-type paint finishes.

6.3.4 Parking Area Signs

A special type of vehicular signage, Parking Area Signs are important parts of the wayfinding system, functioning at the transition point where drivers become pedestrians. The following design features apply to parking area signs:

- Signs within parking lots should be coordinated with any changes to the naming and function of the site signage system.
- Entrances to parking areas should be clearly labeled. A listing of buildings and functions served by the parking should be placed near the entrance point.
- Each major parking area should have a distinct name, derived from its location or function. This will allow the parking areas to be indicated on directories and directional signs.
- The use of special parking signs should be minimized.
- Disabled accessible parking spaces should include post-mounted signs.

- When reserved parking spaces have been assigned, assign the spaces to a category of users instead of to individuals, where appropriate. Individual space assignments are often empty and do not allow for an effective use of parking resources.
- For parking spaces along a roadway, any reserved or controlled spaces should be limited to curb and pavement markings wherever possible.



Recommended parking sign indicating where permit parking begins



Existing parking area and bus shuttle signage

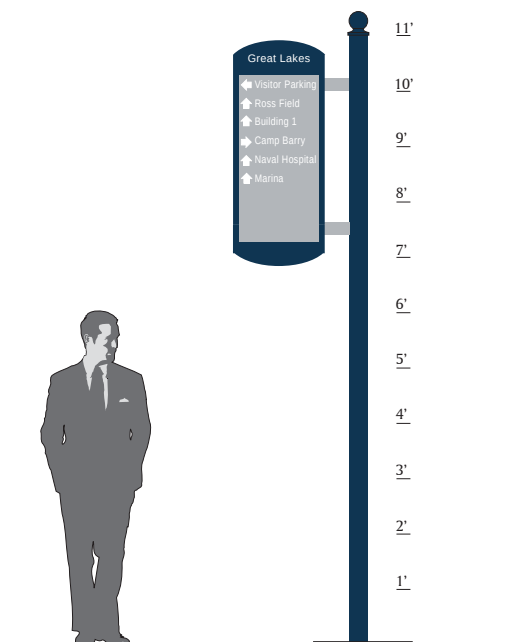


- Post-mounted signs should be limited to designating multi-space parking zones, to decrease the number of reserved signs that are currently at each space. These signs should indicate where the reserved parking starts and where it ends.
- All parking signage should utilize Helvetica Medium lettering in white on a dark brown background.

6.3.5 Pedestrian Directional Signs

Pedestrian Directional Signs, also called wayfinding signs, will provide directions to specific destinations and buildings throughout the sites. The following design features apply to pedestrian wayfinding signs:

- Pedestrian wayfinding signs can have more messages than vehicular signs.



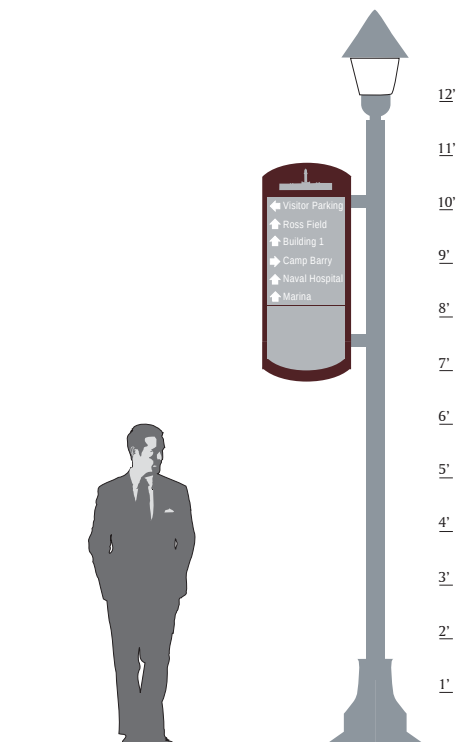
Recommended pedestrian directional sign (typical)

- Messages should direct pedestrians to building entrances.
- Since these signs are viewed from a close distance, better detailing and vandal resistance is called for.

6.3.6 Directories

A special type of pedestrian wayfinding signage, directories are designed to provide overall site maps and lists of destinations for pedestrians. They should be placed at appropriate locations for pedestrians to stop and read the information.

- Directories should be located at key pedestrian locations near parking garage exits, surface lots, and other transition points between cars and pedestrians.



Recommended pedestrian directional sign for the historic district

- Change is inevitable. The directories should have changeable maps, artwork, and name lists.
- Directories would benefit greatly from illumination. Backlighting or picture-lighting of the information panels would make them readily apparent in low-light or night situations.



Recommended pedestrian directory

6.4 Identification Signage

Identification Signage for buildings and structures is a critical element of wayfinding. These signs identify the destinations throughout the Installation through different signage type and design.

6.4.1 Site Building Identification

Site Building Identification signs are ground-mounted “monument” signs located for visibility from surrounding roads and at the approach to the drop-off point or main entrance.

The following design features apply to site building identification signs:

- Identification signs should be integrated with the buildings and the site. Site building identification should be oriented to the primary street and should be visible to drivers.
- If sign illumination is desired, attention should be given to avoid light spillage beyond the sign itself. The minimum amount of lighting should be used as required for safety and comfort.



Existing site building identification signs



- Messages should be simplified to the building name, number and/or street address only. Excessive messages reduce the type size and legibility.
- Messages should be at least 6" tall and 2 feet above grade to allow for landscaping around the signs.
- Typefaces and colors should be coordinated with other signage elements.

6.4.2 Building Identity Signs

Building Identity Signs mark each individual destination throughout an Installation as viewed from each approach. Existing building identity sign systems use both building numbers as well as building names on many structures.

Two primary sign types are used; the following design features apply to the specified type of sign.

BUILDING NUMBER PANELS

- Building number panels should be located on at least two highly visible corners of buildings, determined by the proximity of the primary street that accesses the facility and the primary building entrance.

- The bottom of the building number panel should be located six feet above grade. The signs should be located one foot from the building edge, unless otherwise recommended by the NEPA and Cultural Resources or Historic Preservation manager.
- Building number panels should be bolted to the building.
- The building number signs should be rectangular sheet metal in natural aluminum with colored panels per the Functional District requirements (typical) or dark brown (historic district).
- The building number should be in reflective white lettering, in Helvetica Medium (typical) or Palatino Linotype (historic district).
- Existing, non-uniform signs should be replaced with signs using the new project colors, icons and typefaces.

BUILDING NAME PANELS

- Building name panels are optional; if used, they should be located near or above the building entrances.



Recommended site building identification sign

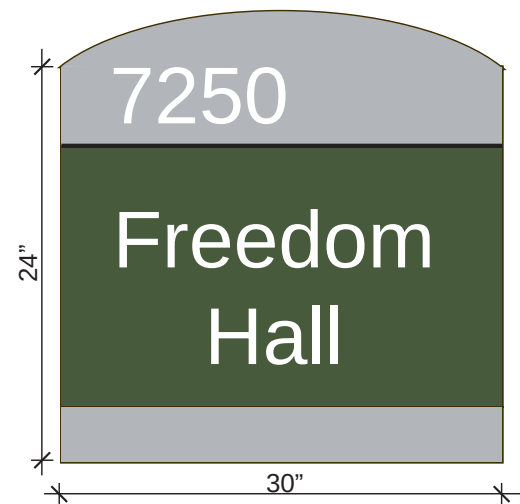


Existing building number and name signage.

- Where site building identification signage is present, building name panels shall not be used.
- The building name panels should be sheet metal in natural aluminum with colored panels per the Functional District requirements (typical), or dark brown (historic district). Both variations include a natural aluminum panel with a clear acrylic coat, upon which the name of the building is provided. Lettering should be reflective white letters, four to five inches in height.
- Limit messages to the primary use of the building only. Shorter messages are better.



Recommended building number panel (historic district).



Recommended building name panel (typical).

6.5 District-Specific Signs

INTERPRETIVE SIGNS

Interpretive signs should be used sparingly to avoid adding visual clutter to the very areas or elements they were designed to highlight. Any new signs in the historic district should match the existing interpretive signs.

BANNERS

The light standards on the primary streets around Ross Field are equipped with banner mounts. These banners emphasize the streets' importance, enliven the roadway experience and inform Installation personnel and visitors about upcoming events.

Banner systems should only be located on primary streets or VIP routes. Additional banner mounts should be installed as needed along the primary streets—Farragut, Decatur, and Dewey Avenues—that already have some banners.



Existing interpretive signage in the historic district

BASE-WIDE GUIDELINES

- Street poles at the entry gates and the primary streets should be equipped with banner mounts similar to those around Ross Field.

ADDITIONAL GUIDELINES FOR THE HISTORIC DISTRICT

- Design and Installation specifications for any sign to be sited within the historic district must be reviewed and approved in writing by the NEPA and Cultural Resources or Historic Preservation manager, as appropriate.
- Banners in the historic district should include text or imagery referring to the NAVSTA Great Lakes Historic District.
- Color schemes, lettering, and graphic schemes should complement the permanent signage and color schemes within the historic district.

SPECIAL EVENTS SIGNS

Signage related to regular events, such as directions to weekly graduations, should be incorporated into permanent vehicular and pedestrian directional signage.



This example of a flat panel sign includes a logo as well as the Installation's signage theme



6.6 Amenity Identity Signs

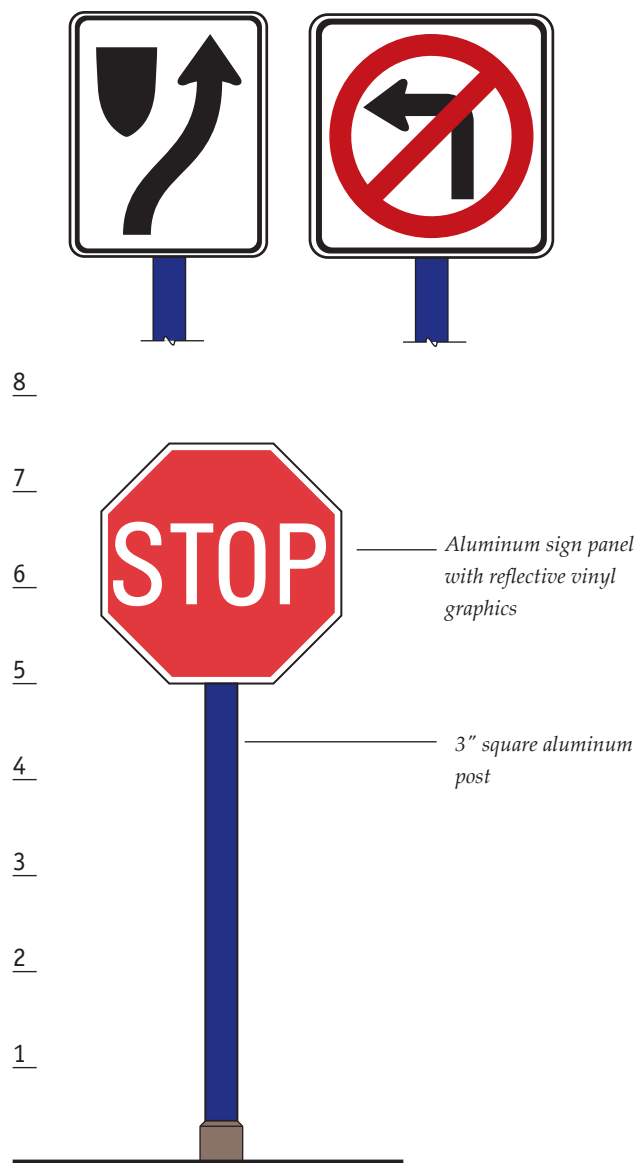
Signs for commercial uses, such as retail stores, restaurants, and non-government functions, should complement the character of the Installation. They should become a part of the overall composition, but not be allowed to dominate the non-commercial signage. To ensure the best integration of commercial signs into the project, detailed commercial signage criteria should be prepared as a separate exercise. The criteria should define the signage requirements in detail, including certain standards of quality and design which are compatible with each Installation's architecture and design vocabulary.

Some basic points which should be required of commercial signs are:

- Messages on Commercial signs shall be limited to the tenant's trade name or logo only.
- Commercial signs should be mounted on panels to reflect appropriate signage themes on the Installation. Appropriate sign types may also include panel signs with dimensional frames mounted flat to the wall surface, simple awnings with graphics, gold-leaf on glass, or projecting blade signs.
- Use of simple lettering compatible with the theme of the Installation is encouraged.
- Natural materials such as painted or natural wood, natural metals, brass or copper, wrought iron, tile, stained glass and stone are encouraged for the signs and their details.
- Signs shall have incandescent lighting only. Typical lighting should be by projecting gooseneck incandescent lights. Internally illuminated sign boxes and individual letters with plastic faces are not allowed.

6.7 Regulatory Signs

Regulatory signs include standard traffic control signage, as well as many messages specific to the Installation, such as restricted or hazardous areas, hours of operation, and deliveries. Although they are primarily functional, the design of regulatory signs should relate to the rest of the system with inexpensive detailing, such as paint colors and typeface selection. Existing signs should be replaced on an on-going basis as the site is redeveloped. Regulatory signs are dictated by the



Traffic signs should use standard Department of Transportation (DOT) sizes, shapes and colors



Federal Highway Administration (FHWA) for travel and parking. Included in this category are speed limit signs, turning and lane usage signs, and warning signs.

TRAFFIC CONTROL SIGNAGE

The following requirements apply to traffic control signage:

- For safety and legal reasons, critical traffic control signage (Stop, Yield, etc.) and identification of hazardous areas should be located by a qualified traffic consultant or civil engineer.
- All regulatory signs at NAVSTA Great Lakes should conform to the standards contained in the latest edition of the Manual on Uniform Traffic Control Devices (MUTCD) by the FHWA. The MUTCD provides federal standards for design, shape, color, dimension, symbol, word message, lettering and location.
- Stop signs should not be placed on a square background.
- Wall-mounted signs may be used where appropriate. These can provide visibility while maintaining sidewalk clearance and reducing damage to the signs.



Existing electronic sign at the health clinic

6.8 Electronic Signs

Electronic signage is a relatively new phenomenon at Navy Installations. To the extent possible, electronic signage should be designed in accordance with the principles outlined in Section 6.4. Site Building Identification in terms of dimensions and materials used for its base. However, manufacturing requirements may dictate specific design features. The following guidelines apply to electronic signs:

- Signage should be rectangular, horizontal in orientation, with the display area generally comprising no more than ten square feet. Pediments or other decorative elements that deviate from the rectangular form should be avoided.
- The sign should rest on a vertical support that should be no more than 48 inches high.
- The frame of the sign and the support should be painted Pantone 7532C (brown) to be consistent with the other auto-oriented signage on the Installation.
- Use of electronic signs should be limited to no more than one per primary gate per Installation, employing them only for the dissemination of key information that is subject to change.
- New electronic signs associated with commercial uses, such as date/time displays common for banks, are not permitted.
- Consistent with the placement of conventional signs, electronic signs should be placed a minimum of six feet from the roadway.



- Light emitting diode (LED) signs should be employed rather than conventional electronic signs, as LED signs consume approximately one-tenth the amount of electricity required to power an incandescent sign. In addition, resolution is higher with LED technology.
- Luminance should be restricted at night to that which is necessary to maintain readability. Excessive luminance can contribute to light pollution at the Installation.
- Electronic signage with streaming text or changing screens should not be placed within the first 100 feet of entry gates or at complex or high-volume intersections, as they have the potential to increase driver distraction, thereby creating a safety hazard.
- Electronic signage should not be positioned where the information load is already high due to conventional signage.
- Electronic signage should not be located where drivers are frequently performing lane changing maneuvers.
- Text streams placed at stop signs should be no longer than ten seconds in duration, to ensure that traffic flow is maintained.
- Double-sided signs should only be installed where the message can be clearly read from sides of the roadway without creating visual obstructions.
- All information, with the exception of the Installation logo, should be located within the electronic panel of the sign.

6.9 Specifications

Specifications for the signage types illustrated in this chapter are provided below. The intent of these specifications is to ensure that directional and building identification signage are continuous and consistent across the base.

For each sign type, features addressed include location, font, color, arrows, number of panels, type of signage system, and estimated costs.

The suggested products, made by Howard Industries, are currently in use at other Navy Installations. Howard Industries' Wordbar system is the anticipated signage system, in keeping with the base's current requirements, although an equivalent may be installed.

Estimated costs have been provided by the North Shore Sign Company, a vendor who has provided signage to the base.

It is the procurer's responsibility to ensure that the specified system meets the material specifications and that the signage system can withstand severe climate conditions.



LOCATION

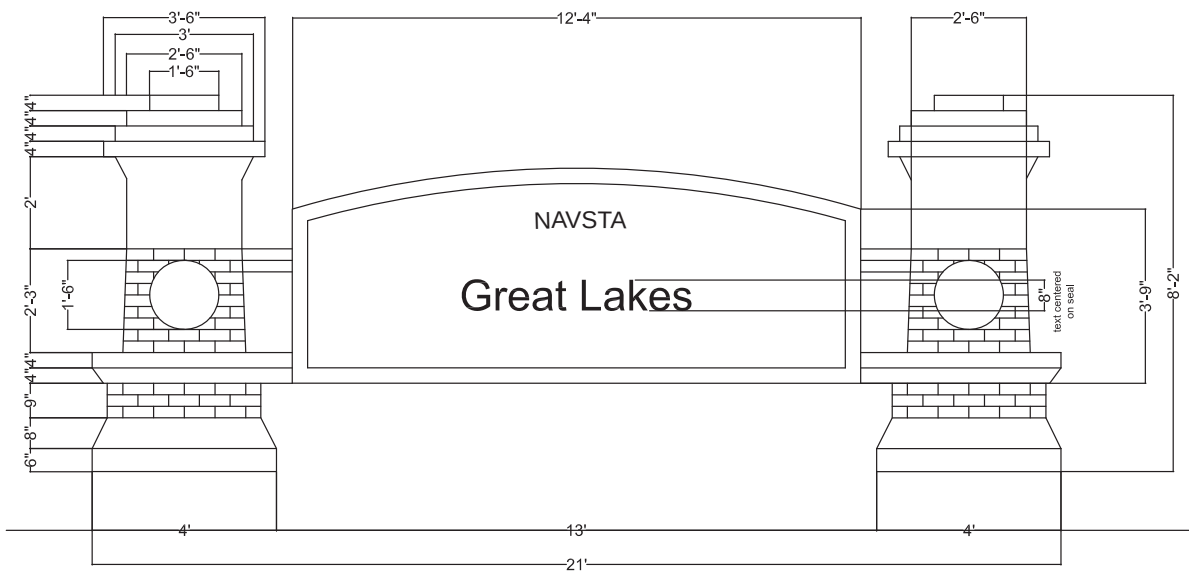
- FONT

- COLOR

- Lettering: Midnight Navy, Pantone 19-4110 TPX.
- Materials

- ## SIGNAGE SYSTEM

Custom design.



6.9.2 Site Entry Identification Sign (Option 2)

LOCATION

- Siting recommended approximately 20' from surrounding public way, marking entry to gates.

FONT

- Upper and lower case Helvetica Medium.
- Capital letter height 3".

COLOR

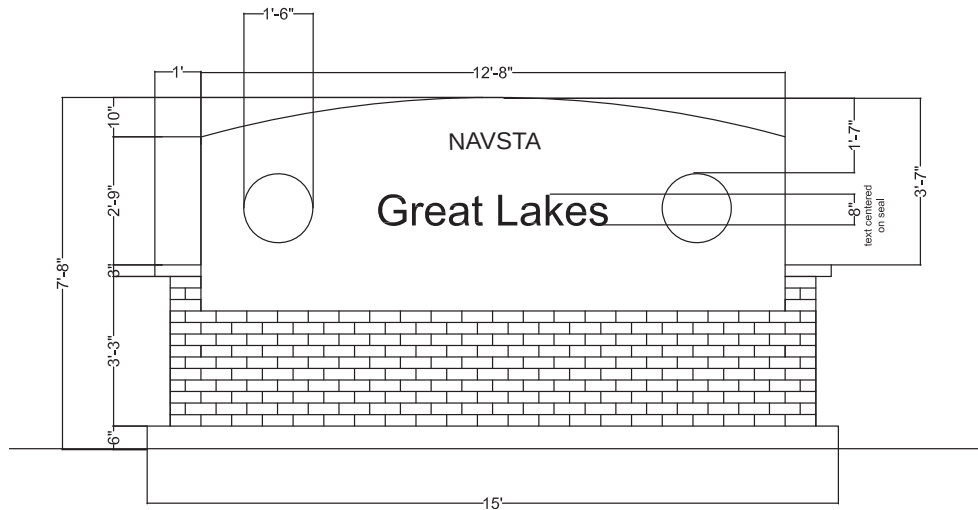
- Lettering: Midnight Navy, Pantone 19-4110 TPX.

MATERIALS

- Concrete base and pillars.
- Red brick to match recommended building materials.
- Natural aluminum panel (match to High Rise, Pantone 15-4101 TPX).
- Logo emblems mounted to aluminum panel.

SIGNAGE SYSTEM

Custom design.



6.9.3 Street Name Identification (Typical)

LOCATION

- One sign at each intersection.
- Site 4' from curb at corner.

FONT

- Upper and lower case Helvetica Medium.
- Capital letter height 3".

COLOR

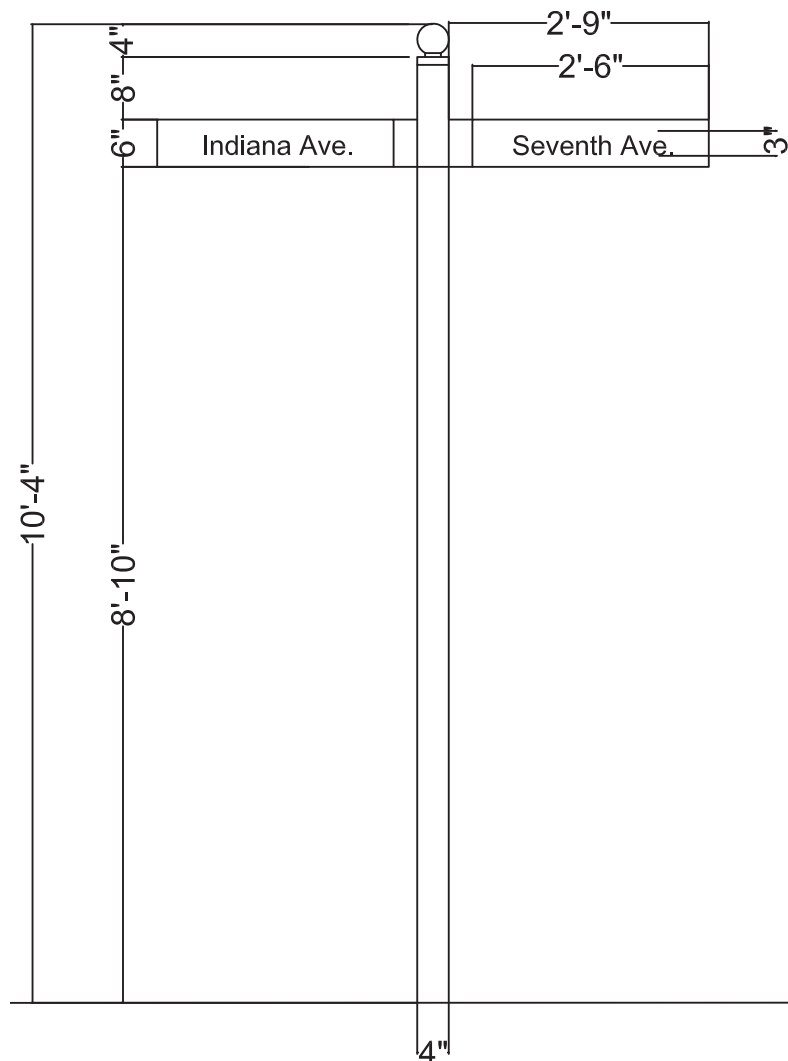
- Sign panels: Classic Blue, Pantone 19-4052 TPX.
- Stripe detail: match to designated District.
- Lettering: reflective white text.

MATERIALS

- Pole: 4" square galvanized pole, with decorative galvanized cap.

SIGNAGE SYSTEM

Sheet metal; 0.080" aluminum panel.



6.9.4 Street Name Identification (Historic District)

LOCATION

- One sign at each intersection.
- Site 4' from curb at corner.

FONT

- Upper and lower case Palatino Linotype.
- Capital letter height 3".

COLOR

- Sign pole and panel surround: Midnight Navy, Pantone 19-4110 TPX.

- Sign panel: Natural aluminum, matched to High Rise, Pantone 15-4101 TPX, with 1" border in Midnight Navy, Pantone 19-4110 TPX, around edge of sign panel.

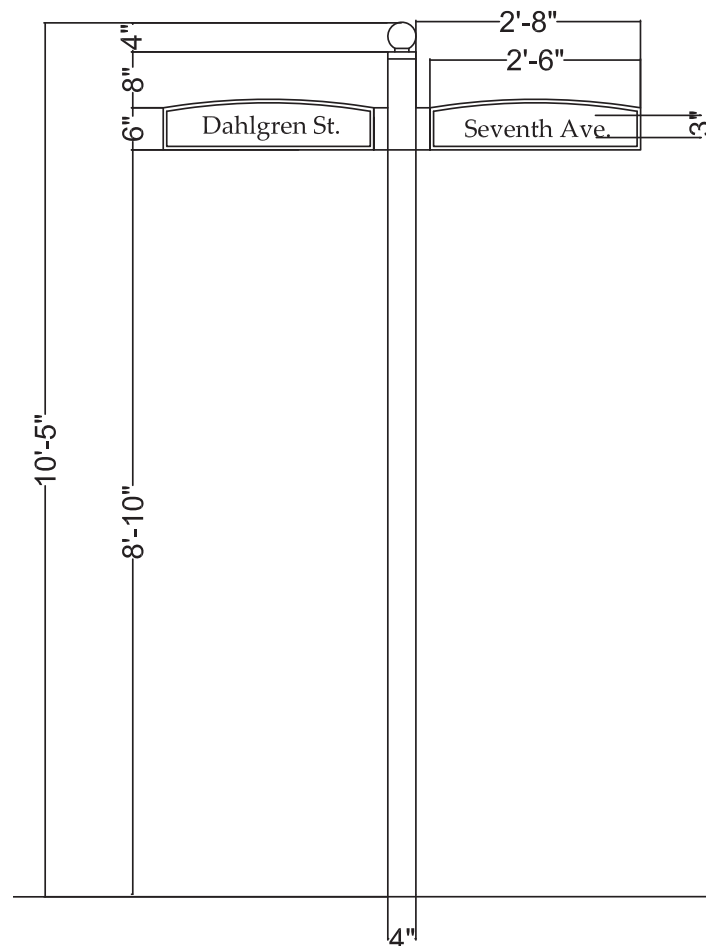
- Lettering: reflective white text.

MATERIALS

- Pole: 4" square galvanized pole, with decorative galvanized cap.

SIGNAGE SYSTEM

- Sheet metal; 0.080" aluminum panel.



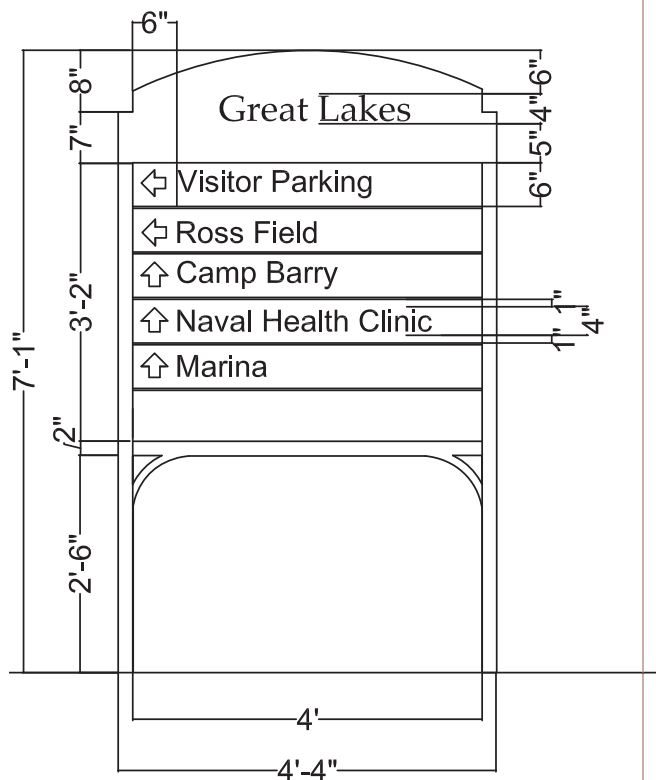
6.9.5 Vehicular Directional Signage

LOCATION

- Right side of road, facing traffic; sign is single-sided.
- Site 100' before an intersection, 6' from curb at closest point.
- Rotate sign 3° toward the roadway to reduce reflectivity.

FONT

- Upper and lower case Palatino Linotype for title.



- Upper and lower case Helvetica Medium for general content.

COLOR

- Sign case: light natural aluminum (match to High Rise, Pantone 15-4101 TPX), separated by 1/4" white rule.
- Sign panel, title: match to designated District.
- Sign panels, directional: Classic Blue, Pantone 19-4052 TPX.
- Lettering and arrows: reflective white text and graphics.

ARROWS

- 3-1/2" wide and high.
- Center arrow graphics in 6" space as designated in diagram.

NUMBER OF PANELS

- Install minimum of three panels and maximum of six panels.
- For signs with fewer than six panels, install matching blank panels at the bottom.

SIGNAGE SYSTEM

- Howard Industries WORDBAR system, with duranodic bronze posts.



6.9.6 Special Parking Area Designations

LOCATION

- At head of angled, head-in, or parallel spaces to designate beginning and end of special parking designation.
- Position to face driver.

FONT

- Upper and lower case Helvetica Medium.
- Capital letter height 2".

COLOR

- Sign panels: brown, Pantone 7532C.
- Lettering: reflective white text and arrows.
- White border around edge of sign panel.

ARROWS

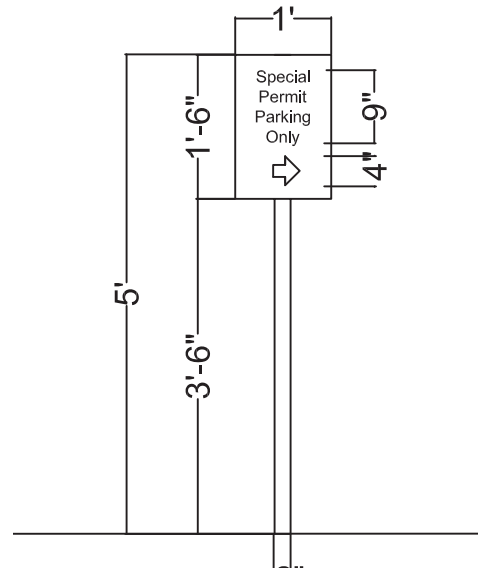
- 2" wide and high.
- Center arrow graphics below message.

MATERIALS

- Pole: 2" square galvanized pole.

SIGNAGE SYSTEM

- Sheet metal; 0.080" aluminum panel.



6.9.7 Pedestrian Directional Signage

LOCATION

- At pedestrian connections adjacent to parking areas, and along walkways to pedestrian destinations; double-sided for visibility from both approaches.
- Where adjacent to drives, site between pedestrian path and building.

FONT

- Upper and lower case Palatino Linotype for title.
- Upper and lower case Helvetica Medium for general content.
- Capital letter height 2.5" for title.
- Capital letter height 2" for general content.

COLOR

- Sign case: light natural aluminum (match to High Rise, Pantone 15-4101 TPX).
- Sign panel, title: match to designated District.
- Sign panels, directional: Classic Blue, Pantone 19-4052 TPX.
- Lettering: reflective white text and arrows

ARROWS

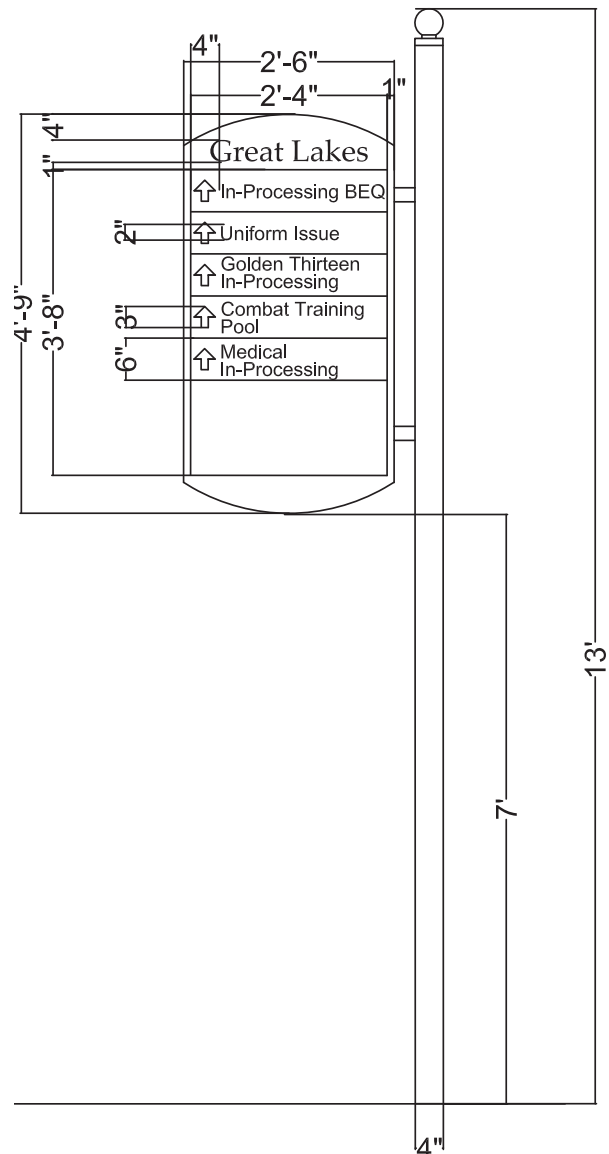
- 3" wide and high
- Center arrow graphics in 4" space as designated in diagram.

NUMBER OF PANELS

- Install minimum of three panels and maximum of seven panels.

SIGNAGE SYSTEM

- Howard Industries WORDBAR system, with duranodic bronze frame.



6.9.8 Pedestrian Directory

LOCATION

- At pedestrian connections adjacent to visitor parking areas and entry from Metro Station.
- Orientation of map shall match direction faced by viewer; north arrow shall be included in graphic, and sign may be double-sided if appropriate.

FONT

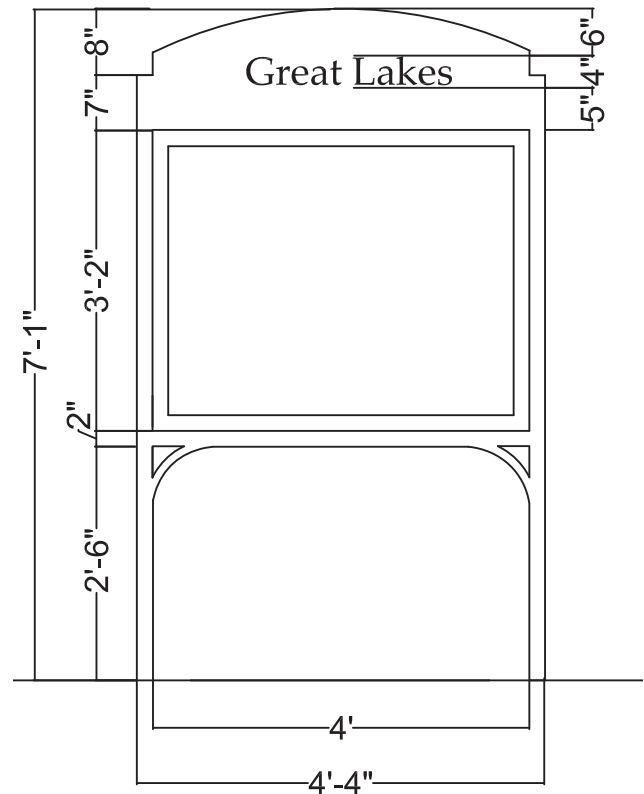
- Upper and lower case Palatino Linotype for title.
- All text on map graphics to be Helvetica Medium; general labels should be 1/4", with headings and area labels of minimum 1/2" capital letters.

COLOR

- Sign case: light natural aluminum (match to High Rise, Pantone 15-4101 TPX).
- Sign panel, title: match to designated District.
- Lettering, title: reflective white text.

SIGNAGE SYSTEM

- Howard Industries HED-100 post and panel system, with duranodic bronze posts and surround.



6.9.9 Site Building Identification Sign

LOCATION

- Adjacent to parking access drives.
- Site parallel to roadway, with text facing roadway.

FONT

- Upper and lower case Helvetica Medium.
- Capital letter height 6" for building name; 4" numerals for building number.

COLOR

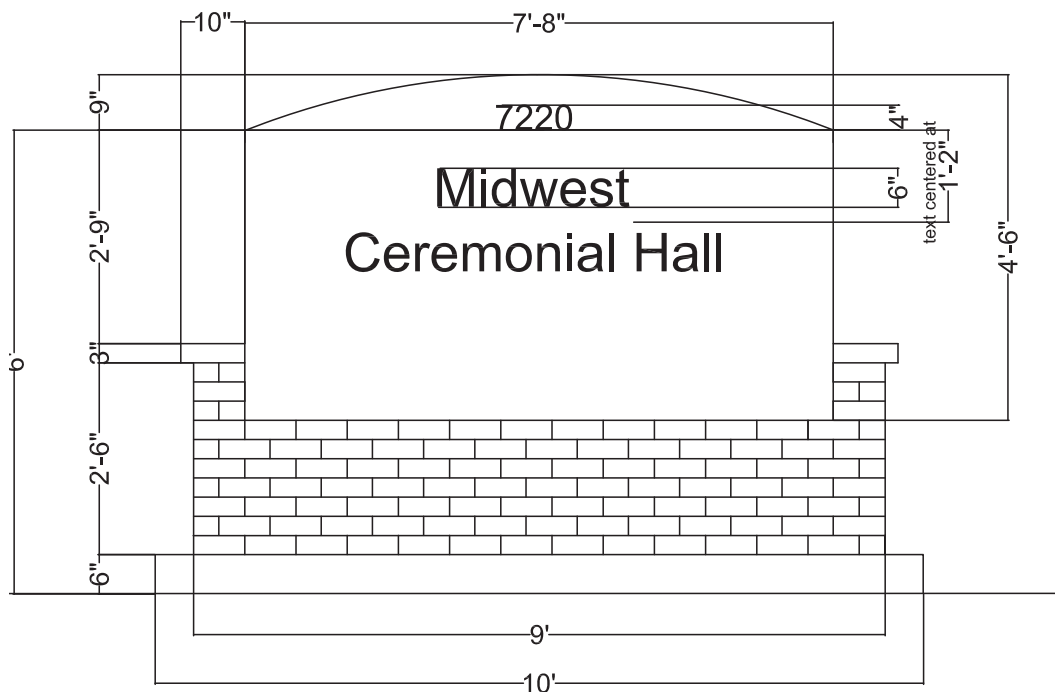
- Lettering: Midnight Navy, Pantone 19-4110 TPX.

MATERIALS

- Concrete base.
- Red brick to match recommended building materials.
- Natural aluminum panel (match to High Rise, Pantone 15-4101 TPX).

SIGNAGE SYSTEM

- Custom design.



6.9.10 Building Number Panel (Typical)

LOCATION

- At two highly visible corners of each building, determined by the proximity of the primary street that accesses the facility and the primary building entrance.
- Install six feet above grade, one foot from the building edge.

FONT

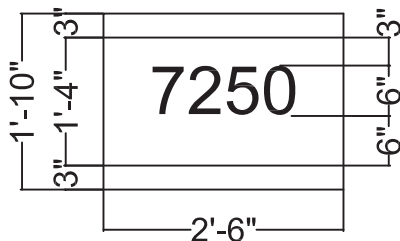
- Building number and name, as applicable: Upper and lower case Helvetica Medium.
- Numeral heights as given on diagram.

COLOR

- Sign panel surround: Natural aluminum, matched to High Rise, Pantone 15-4101 TPX.
- Sign panels, typical: match to designated District.
- Lettering: reflective white text.

SIGNAGE SYSTEM

- Sheet metal; 0.080" aluminum panel, bolted to building.



6.9.11 Building Number Panel (Historic District)

LOCATION

- At two highly visible corners of each building, determined by the proximity of the primary street that accesses the facility and the primary building entrance.
- Install six feet above grade, one foot from the building edge.

FONT

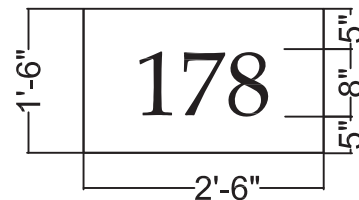
- Upper and lower case Palatino Linotype.
- Numeral heights as given on diagram.

COLOR

- Sign panels: Decadent Chocolate, Pantone 19-1625 TPX.
- Lettering: reflective white text.
- White border around edge of sign panel.

SIGNAGE SYSTEM

- Sheet metal; 0.080" aluminum panel, bolted to building.



6.9.12 Building Name Panel

USE

- Optional, only for use as follows:
 - where room does not permit for site building identification or
 - where it is necessary to identify building names and numbers from approaches other than those adjacent to the main entrance (such as pedestrian approaches).
- Only single-use buildings or buildings known by a single name shall be identified with a Building Name Panel.

LOCATION

- Above primary building entry (or to right of primary entry if room does not permit).
- Dimensions assume a viewing distance under 200'; for distances between 200'-300', increase all dimensions by 33%. Sign panels shall not be used for viewing distances over 300'.

FONT

- Typical: Upper and lower case Helvetica Medium.
- Historic District: Upper and lower case Palatino Linotype.
- Capital letter height 3".

COLOR

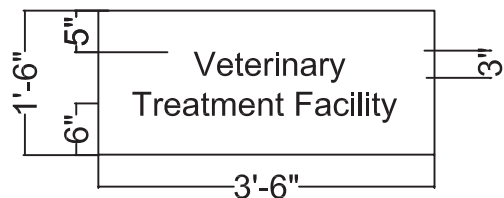
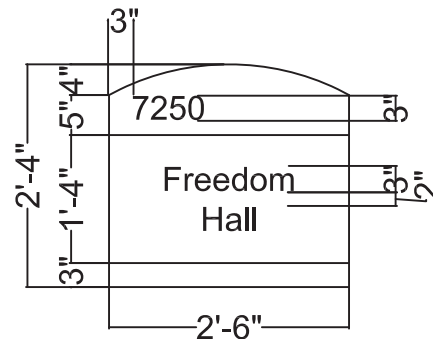
- Sign panels, typical: match to designated District, typical

- Sign panels, Historic District: Decadent Chocolate, Pantone 19-1625 TPX, with white border around edge of sign panel.

- Lettering: reflective white text.

SIGNAGE SYSTEM

- Sheet metal; 0.080" aluminum panel in increments of 9" high for each line of text.



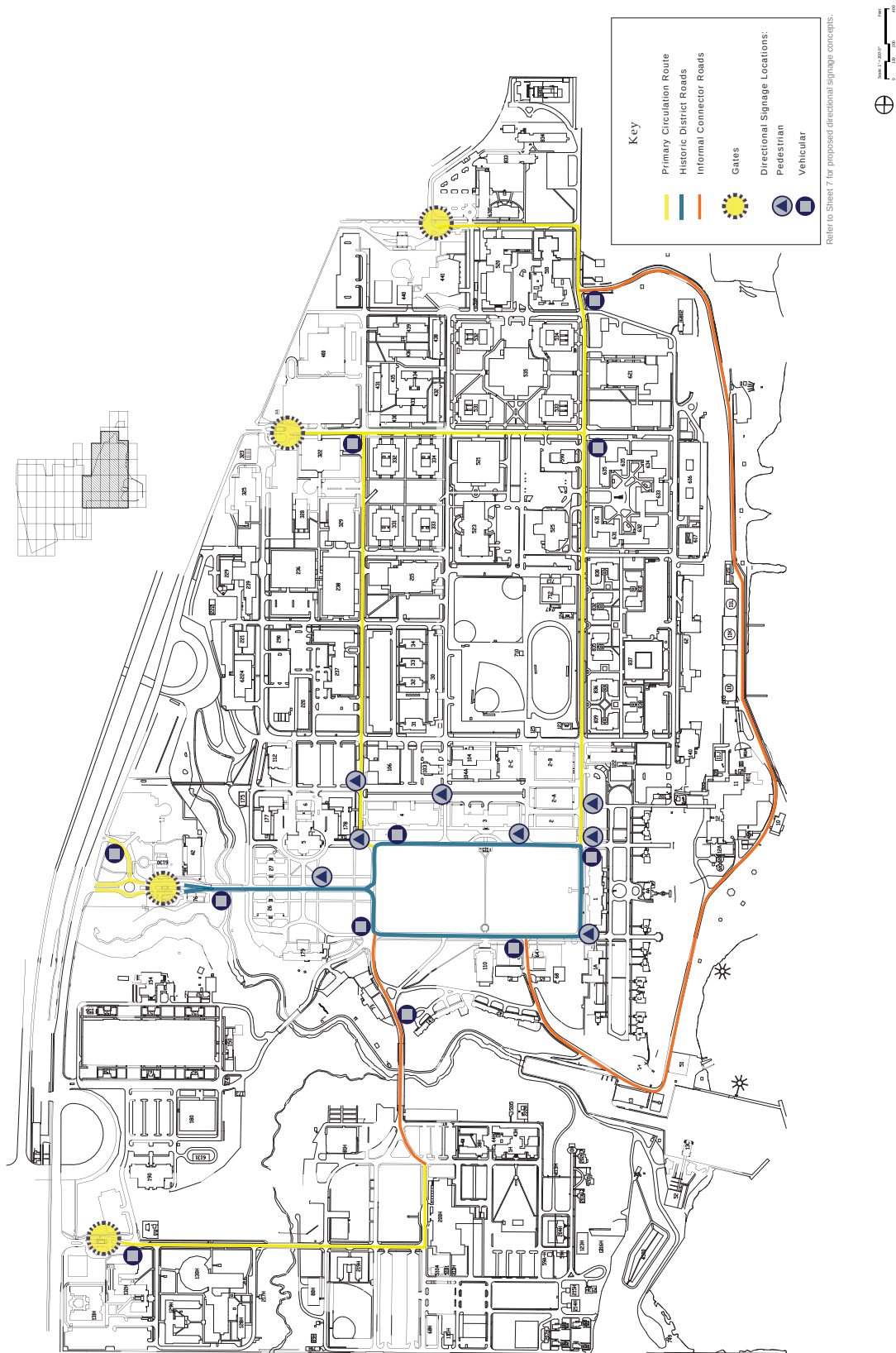


Figure 6.1a Signage Location Plan

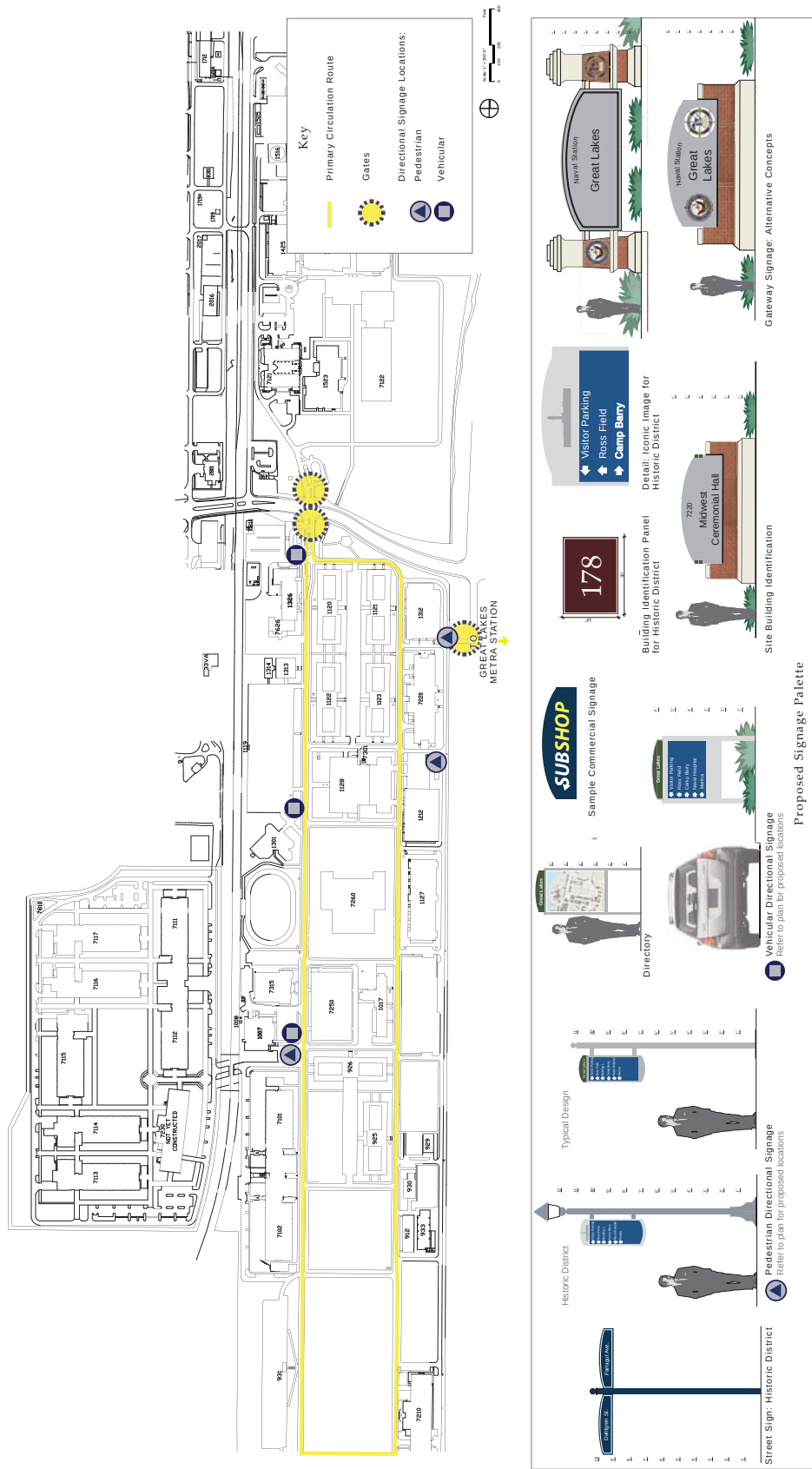


Figure 6.1b Signage Location Plan